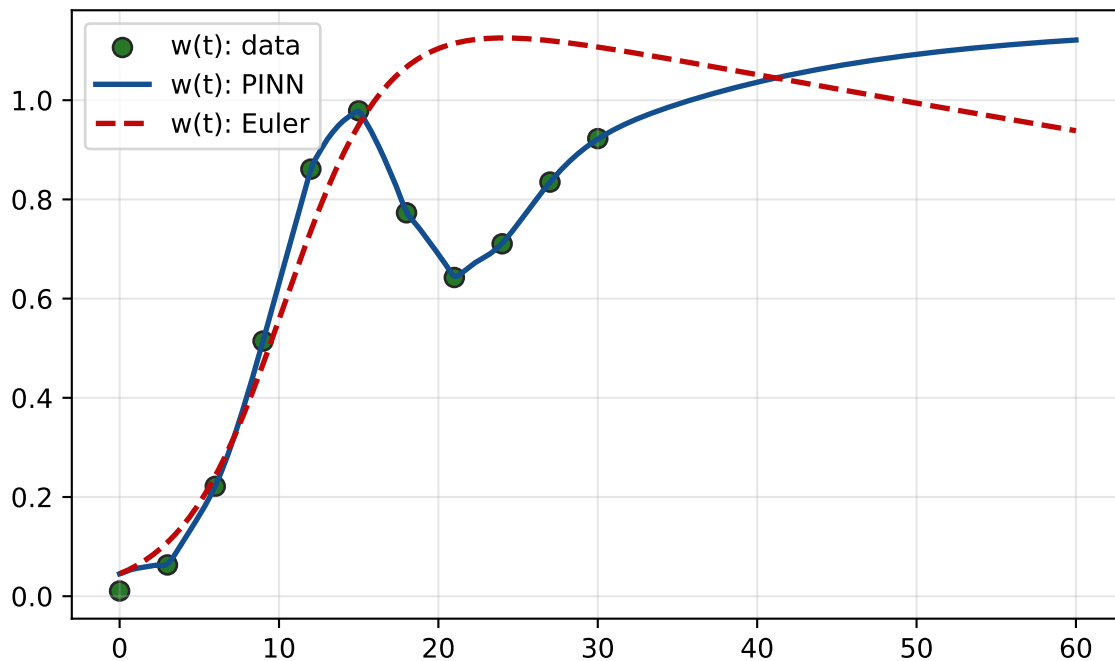
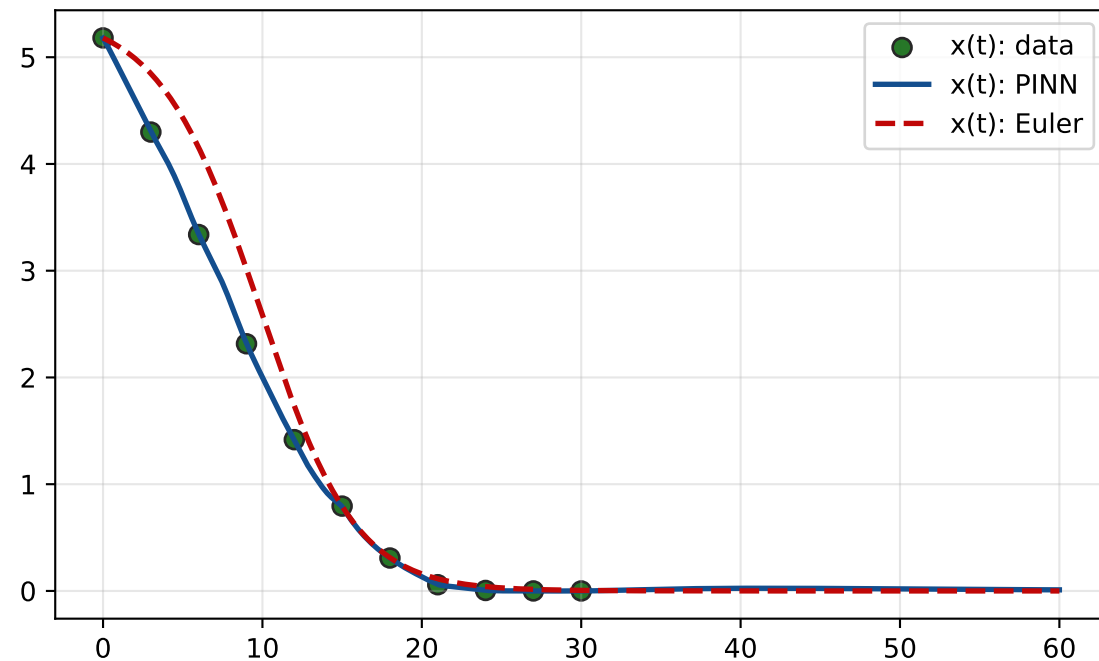


## E8 | PINN vs Euler

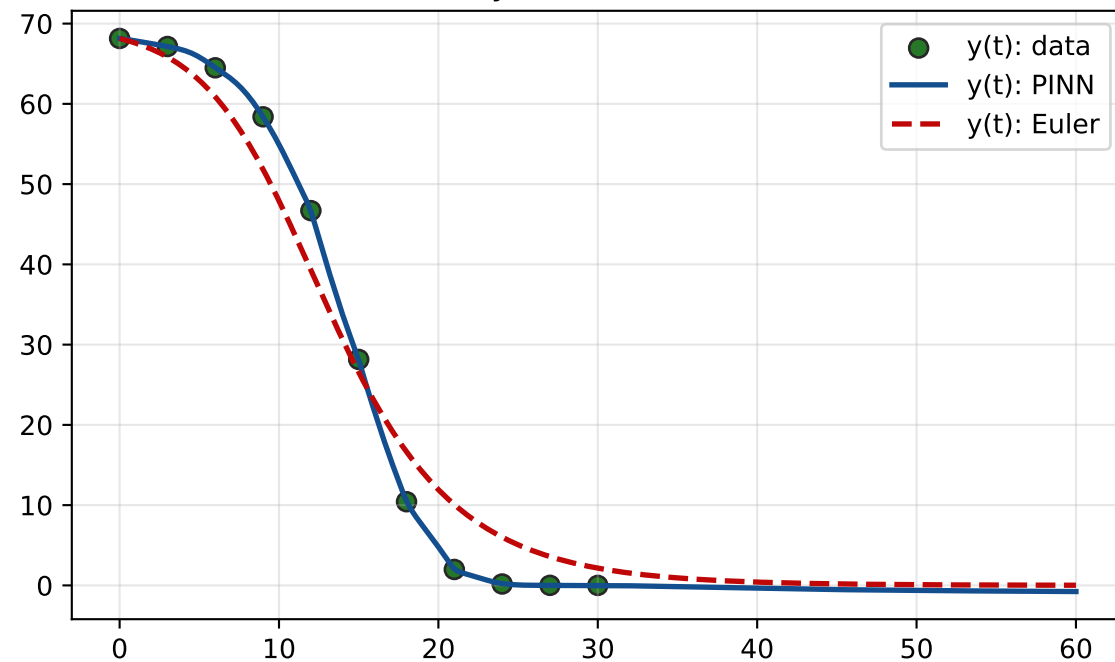
w(t): Biomass



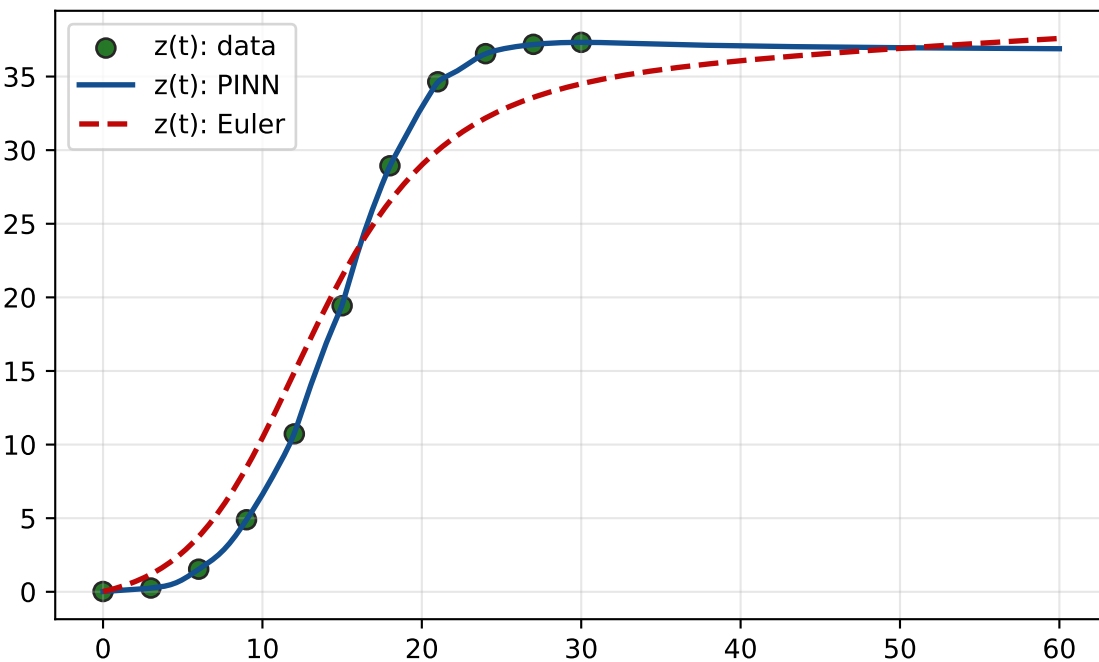
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E8

R2=0.9798, R2\_adj=0.9759, RSS=3.803e+02, AIC=1.120e+02

cond(JTJ)=8.33e+06, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 4.105723e-03 | 1.672630e-03 | 3.389070e-03 | 7.166525e-04  | 7.494793e-03 | 1.892705e-02 |
| p2    | 9.973078e-02 | 2.748388e-01 | 5.568764e-01 | -4.571456e-01 | 6.566072e-01 | 7.187668e-01 |
| p3    | 1.504517e-04 | 3.486040e-03 | 7.063388e-03 | -6.912936e-03 | 7.213839e-03 | 9.658074e-01 |
| p4    | 3.063721e-01 | 3.568190e-01 | 7.229839e-01 | -4.166119e-01 | 1.029356e+00 | 3.960784e-01 |
| p6    | 1.527592e-01 | 9.825221e-02 | 1.990779e-01 | -4.631873e-02 | 3.518371e-01 | 1.285150e-01 |
| p8    | 7.378823e-02 | 4.829530e-02 | 9.785558e-02 | -2.406735e-02 | 1.716438e-01 | 1.350531e-01 |
| p9    | 1.956752e-04 | 1.212694e-02 | 2.457151e-02 | -2.437583e-02 | 2.476718e-02 | 9.872129e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 4.502399e-02 |
| x0 (fixed)     | 5.180769e+00 |
| y0 (fixed)     | 6.814856e+01 |
| z0 (fixed)     | 1.921567e-02 |
| W0 (fixed)     | 0.000000e+00 |